

# Mid-Semester Thesis Report: Dementia Detection from Speech and Text

Oct 5, 2026 · @Ishita

## 1. Objective

This thesis builds and evaluates a system that detects Alzheimer’s disease (AD) from spontaneous speech, starting with transcripts and moving to audio. By mid-semester, a text-based pipeline is working on DementiaBank data and reaches about 91.5% to 93.4% accuracy on a speaker-level held-out split, with caveats listed in Section 6. An earlier transcript-level split gave 96.4%, which probably overstated accuracy because the same speaker could appear in both train and test.

Speech is a cheap, non-invasive signal, and changes in vocabulary, grammar and fluency can appear early in AD. The work so far has four goals:

- Classify each speaker as AD or healthy control (HC) from the transcript of their speech.
- Find which linguistic features carry the most predictive signal.
- Compare handcrafted linguistic features against a simple TF-IDF bag-of-words baseline.
- Prepare the experiments for the second half: audio features, robustness to ASR errors, and generalisation across corpora.

## 2. Dataset

The dataset is 558 transcripts from DementiaBank: 315 AD (56.4%) and 243 HC (43.6%). Access was obtained through a registered DementiaBank account.

| Item           | Detail                                                                  |
|----------------|-------------------------------------------------------------------------|
| Source corpora | Pitt, DePaul, Baycrest, Delaware, Hopkins, Kempler, Lanzi, Lu, VAS, WLS |
| Format         | CHAT (.cha) transcripts                                                 |

| Item          | Detail                                                                                                                                                                       |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Text used     | Participant (PAR) utterances only, read with pylangacq                                                                                                                       |
| Layout        | data/transcripts/AD/ and data/transcripts/HC/                                                                                                                                |
| Total samples | 558 (315 AD, 243 HC)                                                                                                                                                         |
| Holdout split | 80/20 split grouped by speaker, so no participant appears in both train and test. The original transcript-level split had 446 train and 112 test transcripts (63 AD, 49 HC). |

The interviewer's lines are excluded, so the model sees only what the participant said.

### 3. Process

The work follows one pipeline, from raw transcripts to evaluated models.

1. **Data access.** Download the CHAT transcripts of each corpus from DementiaBank.
2. **Consolidation.** Merge the corpora into one folder tree, split into AD and HC.
3. **Parsing.** Read each .cha file with pylangacq and keep only the participant's words.
4. **Feature extraction.** Compute 11 linguistic features per transcript and save them as features.csv (558 rows).
5. **Preprocessing.** Standardise the features, then pass them through a feature-selection step (SelectKBest,  $k = 20$ ).
6. **Training.** Fit Logistic Regression, SVM (RBF) and Random Forest, all with balanced class weights.
7. **Evaluation.** Report speaker-level cross-validation and an 80/20 speaker-level holdout, so the same participant never appears in both train and test, using accuracy, F1 and AUC.
8. **Baseline.** Run a TF-IDF (unigram and bigram) text classifier on the raw transcripts for comparison.
9. **Feature ablation.** Remove one feature at a time, retrain, and record the drop in accuracy.

## 4. Model Explanation

The main model is a Logistic Regression classifier trained on 11 linguistic features, each chosen because the AD literature links it to language decline.

| Feature                      | What it measures                                                                         | Group                   |
|------------------------------|------------------------------------------------------------------------------------------|-------------------------|
| n_words,<br>n_sentences      | Amount of speech produced                                                                | Length                  |
| ttr                          | Share of distinct words (type-token ratio)                                               | Lexical diversity       |
| mattr                        | Type-token ratio averaged over 30-word windows, less sensitive to text length            | Lexical diversity       |
| avg_word_len                 | Mean characters per word                                                                 | Syntactic complexity    |
| avg_sent_len                 | Mean words per sentence                                                                  | Syntactic complexity    |
| filler_count,<br>filler_rate | Fillers such as "um", "uh", "you know"                                                   | Disfluency              |
| repetition_count             | Immediate word repeats such as "the the"                                                 | Disfluency              |
| pronoun_noun_ratio           | Pronouns divided by nouns; expected to rise when speakers cannot retrieve the right noun | Word-finding difficulty |
| verb_noun_ratio              | Verbs divided by nouns; a marker of difficulty retrieving verbs                          | Word-finding difficulty |

Parts of speech come from the NLTK tagger. After standardisation, Logistic Regression estimates the probability of the HC class as a weighted sum of the features passed through a sigmoid:

$$P(y = \text{HC} \mid x) = \frac{1}{1 + e^{-(w^{\text{top}}x + b)}}$$

The weights  $w$  are learned from the training transcripts, with class weights balanced so the larger AD group does not dominate. A transcript is labelled HC when the probability exceeds 0.5. This model was chosen because each weight can be read directly, which

suits the ablation analysis. SVM and Random Forest are trained on the same features as comparison models, and a TF-IDF classifier on raw text serves as the no-feature-engineering baseline.

## 5. Results

With speaker-level splitting, SVM (RBF) reaches about 93.4% holdout accuracy and Logistic Regression about 91.5%, while TF-IDF with SVM reaches about 94.8% in speaker-level cross-validation. These are the numbers to report. They are lower than the 96.4% from the first run, which was probably inflated by speaker leakage.

| Evaluation (speaker-level)             | Model               | Accuracy |
|----------------------------------------|---------------------|----------|
| Handcrafted features, holdout          | Logistic Regression | ~91.5%   |
| Handcrafted features, holdout          | SVM (RBF)           | ~93.4%   |
| TF-IDF text baseline, cross-validation | SVM (RBF)           | ~94.8%   |

F1, AUC and the Random Forest result for the speaker-level runs are not yet recorded here.

### Earlier transcript-level results

The table and figures below come from the first run, where splits were made per transcript file, so one speaker could appear in both train and test. They are kept for comparison and should not be reported as final. In this run the best result was Logistic Regression on the 80/20 holdout: 96.4% accuracy, F1 0.961 and AUC 0.963 on 112 transcripts. F1 and AUC treat HC as the positive class.

| Experiment                      | Model               | Accuracy | F1    | AUC   |
|---------------------------------|---------------------|----------|-------|-------|
| Handcrafted features, 5-fold CV | Logistic Regression | 92.7%    | 0.918 | 0.969 |
| Handcrafted features, 5-fold CV | SVM (RBF)           | 91.6%    | 0.908 | 0.973 |
| Handcrafted features, 5-fold CV | Random Forest       | 91.0%    | 0.900 | 0.962 |
| TF-IDF text baseline, 5-fold CV | Logistic Regression | 94.1%    | 0.936 | 0.983 |
| TF-IDF text baseline, 5-fold CV | SVM (RBF)           | 94.6%    | 0.941 | 0.982 |

| Experiment                    | Model               | Accuracy | F1    | AUC   |
|-------------------------------|---------------------|----------|-------|-------|
| Handcrafted features, holdout | Logistic Regression | 96.4%    | 0.961 | 0.963 |
| Handcrafted features, holdout | SVM (RBF)           | 94.6%    | 0.942 | 0.992 |
| Handcrafted features, holdout | Random Forest       | 93.8%    | 0.932 | 0.993 |

On the holdout set, the Logistic Regression model labelled all 49 HC transcripts correctly and 59 of 63 AD transcripts correctly. The 4 errors were AD speakers predicted as HC.

The TF-IDF baseline beat the handcrafted features in cross-validation (94.6% against 92.7%). With only 11 features, the handcrafted set captures less than the full vocabulary does, so it may be partly picking up corpus-specific wording.

## Feature ablation

Removing one feature at a time shows which ones the model leans on most. This analysis used the original transcript-level split and should be rerun with speaker-level splits. The impact is the relative drop in holdout accuracy.

| Removed feature    | Impact on accuracy |
|--------------------|--------------------|
| verb_noun_ratio    | 9.3%               |
| pronoun_noun_ratio | 8.3%               |
| avg_sent_len       | 3.7%               |

Both word-finding ratios rank above every length, diversity and disfluency feature. This agrees with the literature on noun and verb retrieval difficulty in AD. The ablation measures importance, not direction, so the sign of each effect still needs to be confirmed from the model coefficients.

## 6. Limitations of the Current Results

The speaker-level figures are the ones to report, and the original 96.4% is superseded. These points still limit the conclusions:

- **Speaker leakage, now addressed.** The first run split by transcript file, so one participant's visits could land in both train and test. The pipeline now groups files by speaker ID and the speaker-level results use that split. The ablation and ASR scripts

still need to be rerun the same way.

- **No like-for-like benchmark.** The 80-90% ADRess figure comes from a different, balanced and demographically matched dataset with a fixed test set. It is not directly comparable to a mixed 558-sample set.
- **Small test set.** One holdout sample is worth about 0.9 percentage points, so gaps of 1-2% between models are within noise. Repeated CV with confidence intervals is needed.
- **Feature selection has no effect yet.** SelectKBest is set to  $k = 20$  but only 11 features exist, so every feature is kept.
- **ASR experiment is a placeholder.** The current script adds Gaussian noise to the feature values, not real transcription errors, so it cannot support a robustness claim yet.
- **Scripts not yet run.** The ASR, BERT-embedding and cross-corpus scripts are written but their results are not yet validated.

## 7. Future Work

The second half of the thesis adds audio, tests robustness to transcription errors, and tightens the evaluation.

1. **Audio features from WAV files.** Download the matching recordings from DementiaBank media, convert MP3 to 16 kHz mono WAV with ffmpeg, and pair each file with its transcript by filename. The pipeline already extracts pause count and ratio, speech-rate proxy, pitch (F0) mean and spread, 13 MFCCs (mean and spread), zero-crossing rate and spectral centroid. Audio captures hesitation and prosody that text cannot, so the aim is to compare text-only, audio-only and fused models and see whether fusion beats the speaker-level text results.
2. **Robustness to ASR errors.** Transcribe the audio with an ASR system such as Whisper, extract features from those transcripts, and compare accuracy against manual transcripts. This replaces the current noise-on-features placeholder, and accuracy can then be plotted against word error rate.
3. **Wav2Vec2 audio embeddings.** The BERT sentence-embedding baseline for text is already done. The remaining step is to extract Wav2Vec2 embeddings from the audio as a deep-learning comparison against the handcrafted acoustic features, and to test them alone and fused with the text models.
4. **Cross-corpus evaluation.** Train on one corpus, test on another (for example Pitt to DePaul) to measure generalisation and rule out a corpus confound.

5. **More rigorous evaluation.** Keep the speaker-level grouped splits and add repeated cross-validation with confidence intervals and per-corpus breakdowns. Fix the feature-selection setting, and report model coefficients to confirm the direction of each feature's effect.